Molecular and cytogenetic analysis of repetitive DNA in pea (*Pisum sativum* L.)

Pavel Neumann, Marcela Nouzová, and Jiří Macas

Abstract: A set of pea DNA sequences representing the most abundant genomic repeats was obtained by combining several approaches. Dispersed repeats were isolated by screening a short-insert genomic library using genomic DNA as a probe. Thirty-two clones ranging from 149 to 2961 bp in size and from 1000 to 39 000/1C in their copy number were sequenced and further characterized. Fourteen clones were identified as retrotransposon-like sequences, based on their homologies to known elements. Fluorescence in situ hybridization using clones of reverse transcriptase and integrase coding sequences as probes revealed that corresponding retroelements were scattered along all pea chromosomes. Two novel families of tandem repeats, named PisTR-A and PisTR-B, were isolated by screening a genomic DNA library with Cot-1 DNA and by employing genomic self-priming PCR, respectively. PisTR-A repeats are 211–212 bp long, their abundance is 2×10^4 copies/1C, and they are partially clustered in a secondary constriction of one chromosome pair with the rest of their copies dispersed on all chromosomes. PisTR-B sequences are of similar abundance (10^4 copies/1C) but differ from the "A" family in their monomer length (50 bp), high A/T content, and chromosomal localization in a limited number of discrete bands. These bands are located mainly in (sub)telomeric and pericentromeric regions, and their patterns, together with chromosome morphology, allow discrimination of all chromosome types within the pea karyotype. Whereas both tandem repeat families are mostly specific to the genus *Pisum*, many of the dispersed repeats were detected in other legume species, mainly those in the genus *Vicia*.

Key words: repetitive DNA, plant genome, retroelements, satellite DNA, *Pisum sativum*.

Résumé : Un ensemble de séquences d'ADN représentant les séquences génomiques répétées les plus abondantes a été obtenu par une combinaison d'approches. Des séquences dispersées ont été isolées en criblant une banque génomique constituée d'inserts de petite taille en utilisant comme sonde l'ADN génomique. Trente-trois clones possédant des inserts dont la taille variait de 149 pb à 2961 pb et dont le nombre de copies variait entre 1000 et 39 000 copies/1C ont été séquencés et caractérisés plus avant. Quatorze clones ressemblaient à des rétrotransposons sur la base de leur homologie à des éléments connus. L'hybridation in situ en fluorescence avec pour sonde les régions codantes de la transcriptase inverse ou de l'intégrase a révélé que les rétroéléments correspondants sont distribués sur l'ensemble des chromosomes du pois. Deux nouvelles familles de séquences répétées en tandem, PisTR-A et PisTR-B, ont été isolées en criblant une banque génomique préparée à partir de l'ADN de Cot-1 et en employant la technique d'amplification PCR auto-amorcée (« self-priming PCR »), respectivement. Les séquences PisTR-A mesurent 211–212 pb, sont présentes en 2×10^4 copies/1C et sont en partie groupées au niveau d'une constriction secondaire d'une des paires de chromosomes, les autres copies étant distribuées sur l'ensemble des autres chromosomes. Les séquences PisTR-B montrent une abondance semblable (10^4 copies/1C) mais diffèrent de la famille "A" de par la taille du monomère (50 pb), le fort contenu en A/T et leur localisation à un nombre limité de bandes discrètes. Ces bandes sont principalement localisées au niveau des régions (sub)télomériques et péricentromériques. De plus, les motifs ainsi produits et la morphologie des chromosomes permettent d'identifier tous les types de chromosomes au sein du caryotype du pois. Tandis que les deux familles de séquences répétées en tandem étaient largement spécifiques au genre *Pisum*, plusieurs des séquences répétées dispersées étaient détectées chez d'autres légumineuses, principalement au sein du genre *Vicia*.

Mots clés : ADN répété, génome de plante, rétroéléments, ADN satellite, *Pisum sativum*.

[Traduit par la Rédaction]

Received January 10, 2001. Accepted April 24, 2001. Published on the NRC Research Press Web site at http://genome.nrc.ca on July 27, 2001.

Corresponding Editor: G. Jenkins.

P. Neumann, M. Nouzová, and J. Macas.[1] Institute of Plant Molecular Biology, Laboratory of Molecular Cytogenetics, Branišovská 31, České Budějovice, CZ-37005, Czech Republic.

[1]Corresponding author (e-mail: macas@umbr.cas.cz).

 DOI: 10.1139/gen-44-4-716 © 2001 NRC Canada

Introduction

The majority of the pea nuclear genome (2*n* = 14, 1C = 4.425 pg; Bennett and Leitch 1997) is composed of repetitive sequences. They are extensively interspersed within the genome and their frequencies range from 100 to 10 000 or more copies per haploid genome, as judged from the reassociation kinetics of pea DNA (Murray et al. 1978, 1981). Although some studies have estimated that the proportion of repetitive DNA in the pea genome is as high as 97% (Murray et al. 1981), only a few repeat families have been isolated and characterized so far. Tandem repeats are represented by 45S and 5S rRNA genes, both with about 4000 copies per haploid genome (Ingle et al. 1975; Ellis et al. 1988), and by clone c41, which has not been fully characterized with respect to its copy number and molecular organization (Ellis et al. 1990). In addition, telomeric repeats originally isolated from *Arabidopsis thaliana* (Richards and Ausubel 1988) were detected in terminal regions of pea chromosomes using fluorescence in situ hybridization (FISH) (Simpson et al. 1990). Known dispersed repeats include the Ty3-*gypsy*-like retrotransposon *Cyclops*, with about 5000 copies per haploid genome (Chavanne et al. 1998), and the Ty1-*copia*-like element *PDR1*, with about 200 copies per haploid genome (Lee et al. 1990; Vershinin and Ellis 1999), that was also successfully used as a codominant genetic marker (Flavell et al. 1998). The presence of abundant and diverse retroelements of both the Ty1-*copia* and Ty3-*gypsy* classes was proposed as a result of studies using PCR with degenerate primers derived from conserved regions of the respective retrotransposon types (Flavell et al. 1992; Suoniemi et al. 1998). Pearce et al. (2000) used this approach to isolate a heterogeneous population of Ty1-*copia*-like elements, two of which were then employed to estimate phylogenetic relationships among different *Pisum* lineages, based on their insertional polymorphisms, which had been detected using a SSAP (sequence-specific amplification polymorphism) method. That at least some of the pea retroelements might be still active, is suggested by the detection of transcripts of conserved reverse transcriptase sequences under protoplasting and fungal-elicitor treatment conditions (Kato et al. 1999).

This work was aimed at isolating a more representative set of the most abundant repeats, to get a better understanding of the composition and molecular organization of this fraction of the pea genome. An emphasis was given to obtaining and characterizing a large number of relatively short sequences that could serve for future isolation of full-length clones of corresponding elements or as molecular probes for genetic or taxonomic studies. Different approaches were used to isolate repetitive sequences, based on their genomic organization. The screening of a "shotgun" genomic library using total genomic DNA as a probe was mainly aimed at isolating dispersed repeats. The other techniques were directed toward isolating highly amplified tandem repeats (satellite DNAs), and included screening the genomic library using Cot-1 DNA as a probe and the cloning of tandem repeats using genomic self-priming PCR (GSP-PCR). All newly isolated sequences were characterized with respect to their nucleotide sequence composition, genomic organization, abundance, and distribution among other legume species.

Materials and methods

Plant material

Seeds of the pea (*Pisum sativum* L.) cultivar Bohatýr were obtained from Osiva Boršov, Czech Republic. Seeds of *Vigna unguiculata* L. were provided by Prof. W.J. Broughton (University of Geneva, Switzerland). Seeds of other Viciaceae species were received from the legume seed bank at Agritec, Šumperk, Czech Republic, or from Institut für Pflanzengenetik und Kulturpflanzenforschung (IPK), Gatersleben, Germany. Seeds of *Arabidopsis thaliana* L. were obtained from Dr. M. Ondřej (Institute of Plant Molecular Biology, Czech Republic).

Cloning procedures

Enzymes, the plasmid vector pBluescript II SK+, *Escherichia coli* strain XL-1 Blue MRF′, X-gal (5-bromo-4-chloro-3-indolyl-β-D-galactopyranoside), and IPTG (isopropyl β-D-thiogalactopyranoside) were supplied by Stratagene (La Jolla, Calif.). Unless otherwise specified, all DNA and bacterial manipulations were performed according to Ausubel et al. (1991). Plant genomic DNA was extracted from young leaves as described by Dellaporta et al. (1983). Concentrations of both genomic and linearized plasmid DNAs were measured using PicoGreen dye (Molecular Probes, Engene, Ore.), according to the manufacturer's protocols.

A pea genomic library was constructed by digesting total DNA with the restriction endonuclease *Taq*I and cloning dephosphorylated DNA into *Bsp*106-cut plasmid pBluescript II SK+. Dephosphorylation of genomic DNA ensured ligation of only single fragments into the plasmid vector, thus avoiding the generation of chimeric clones. Following the transformation of *E. coli* XL-1 Blue MRF′, colonies containing recombinant plasmids were identified using blue–white selection on Lennox L agar medium supplemented with X-gal and IPTG. White colonies (6400) were gridded onto hybridization membranes (Hybond N+; Amersham, England) and screened with pea genomic DNA as a probe. Alternatively, the rapidly reassociating fraction (Cot value = 1) of the pea DNA was used as a probe for screening of selected clones. Cot-1 DNA was prepared by the denaturation and reassociation of sonicated DNA (400–500 bp, on average) in 0.12 M phosphate buffer at 60°C and purification of the double-stranded DNA (dsDNA) fraction using a hydroxyapatite column (DNA Grade Bio-Gel HTP, Bio-Rad, Hercules, Calif.). The hybridizations were done using the AlkPhos Direct labeling and detection kit (Amersham), as recommended by the manufacturer, employing stringent hybridization and washing temperatures (63°C).

Cloning of full-length PisTR-A repeats was accomplished using PCR with the primers F5/1 (5′-ACT CTG CTG GAG ATA ACA ATC G-3′) and F5/2 (5′-ATT CGT ATT CCT TGT GGA ATC G-3′), which were designed according to the corresponding partial clone obtained from the screening of the genomic library. Reactions were run for 30 cycles of 30 s at 94°C, 50 s at 55°C, and 1 min at 72°C; preceded by initial denaturation (3 min at 94°C) and followed by a final extension step (10 min at 72°C). The PCR cocktail included 1× PCR buffer, 0.2 mM dNTPs, 0.2 μM primers, 1.5 mM $MgCl_2$, 4 U/100 μL of Taq polymerase (Promega, Madison, Wis.), and 1 ng of genomic DNA as template. Amplified fragments were cloned using the TOPO TA cloning kit (Invitrogen, Groningen, The Netherlands).

GSP-PCR

GSP-PCR was performed according to Macas et al. (2000), using restriction enzyme digested genomic DNA as primer. The following endonucleases were tested: *Alu*I, *Mbo*I, *Rsa*I, *Taq*I, *Tru*9I, *Bam*HI, *Bgl*II, *Dra*I, *Hinc*II, *Hin*dIII, *Eco*RI, *Eco*RV, and *Kpn*I. Following the reaction, the resulting products were redigested with the respective restrictase and visualized on 8% native polyacrylamide

© 2001 NRC Canada

gels stained with 10 000× diluted SybrGreen (Molecular Probes). Cloning of *Tru*9I-digested products was done by filling-in, and ligation into *Eco*RV-digested pBluescript II SK+.

Sequence analysis

DNA sequencing was done using the dideoxy-mediated chain-termination method (Sanger et al. 1977). Homology searches were performed in the GenBank (release 120) and EMBL (release 64) databases, using the BLASTN or BLASTX 2.0.2 (Altschul et al. 1997) and FASTA 3.2 (Pearson and Lipman 1988) programs. The sequences were further analyzed using PC/Gene 6.60 (IntelliGenetics, Mountain View, Calif.) and Dot-plot 3.0 (Ramin Nakisa, Oxford University).

Southern and dot-blot hybridizations

Southern blots were prepared by digesting 1-μg aliquots of genomic DNA with selected restriction endonucleases, separating the digests on 1 or 1.5% agarose gels, and blotting them onto Hybond N+ membranes by capillary transfer. For genomic copy number estimations, dilutions of pea genomic DNA (50–10^5 copies of the haploid genome) and insert-containing linearized plasmids (10^7–10^9 copies) were quantitatively dot-blotted on Hybond N+ membranes. Pea repeats in other legume species were detected using dot-blots containing 10^5 genome equivalents of the respective species' DNA and 10^8 copies of cloned repeats as controls. The amounts of blotted DNAs were calculated based on genome or insert sizes and assuming that 1 pg of dsDNA was equal to 9.65×10^8 bp (Bennett and Smith 1976).

Hybridization probes were prepared by labeling PCR-amplified plasmid inserts that had been digested with suitable restrictases and gel-purified to remove the surrounding polylinker sequences. Probe labeling and hybridizations were done as described for the genomic libraries, using 63°C as the temperature for hybridization and washing for all probes except PisTR-B, which, owing to its high A/T content was hybridized and washed at 55°C. Hybridization signals were recorded using a Storm scanner, and their intensities quantified using ImageQuant software (Molecular Dynamics, Sunnyvale, Calif.).

FISH

All FISH experiments were performed on isolated chromosomes prepared as described by Gualberti et al. (1996); chromosomes were centrifuged onto slides using a Hettich centrifuge equipped with cytospin chambers. During PCR reactions, hybridization probes for detecting newly isolated pea repeats and for 5S rRNA genes were labeled with biotin-16-dUTP (Boehringer Mannheim, Mannheim, Germany) or Alexa Fluor 568-5-dUTP (Molecular Probes), using T3 and T7 primers and the corresponding plasmid clone as template. The probe Ver17 containing 25S and 18S rRNA genes (Yakura and Tanifuji 1983) was labeled with biotin-16-dUTP using random priming. Chromosome denaturation was carried out in 1× PCR buffer (Promega) supplemented with 4 mM $MgCl_2$ for 3 min at 94°C, and was followed by dehydration using an ice-cold ethanol series. Probe denaturation was done in a hybridization mix lacking SSC (1× SSC: 0.15 M NaCl, 0.015 M sodium citrate) for 15 min at 75°C. Then, the mix was chilled on ice and 20× SSC was added, to obtain a complete hybridization solution (2× SSC, 50% formamide, 10% dextran sulfate, 125 ng/μL of sheared calf thymus DNA, 0.125% SDS (sodium dodecyl sulfate), and 0.025 ng/μL of biotin-labeled probe or 0.5 ng/μL of Alexa Fluor 568 labeled probe). Hybridization was done overnight at 37°C, and post-hybridization washes were done in 2× SSC at 42°C for 5 min and in 50% formamide in 2× SSC at 42°C for 10 min. These conditions were used for all probes except PisTR-B, which was hybridized at 28°C and washed at 32°C. Detection of biotinylated probes was done as described by Leitch et al. (1994) using fluorescein-avidin DN and biotinylated anti-avidin D (Vector, Houston, Tex.). The probes labeled with Alexa Fluor 568 were visualized directly. Chromosomes were counterstained with DAPI (4′,6-diamidino-2-phenylindole) and examined using a Nikon Eclipse 600 microscope equipped with appropriate filter sets. The images were captured with a CCD (charge coupled device) camera and analyzed using Lucia software (Laboratory Imaging, Prague, Czech Republic).

Double-FISH experiments using PisTR-B as one of the probes were performed in two steps. The first step included chromosome denaturation and hybridization with the first probe at 37°C. Following stringent washes at 42°C, slides were dehydrated using the ethanol series, air-dried, and used for hybridization with PisTR-B as described above. This arrangement ensured the hybridization of two probes differing in their melting temperature, without compromising their specificity and sensitivity.

PCR with rDNA and PisTR-A primers

The following primers were designed to detect mutual orientation of rDNA and PisTR-A sequences in the pea genome (see also Fig. 4A). Primers rDNA1 (5′-AAA CGG CTA CCA CAT CCA AG -3′), rDNA2 (5′-CTA AAC CCA GCT CAC GTT CC-3′), rDNA3 (5′-GCC ACG ATC CAC TGA GAT TC-3′), rDNA4 (5′-TTA TGA GCC ATT CGC AGT TTC-3′), JOE1 (5′-TGT GTA GGC CAA GGT TTT GTC-3′), and RS (5′-CGA AAT GCT CCG AAA CTT TGC AGA TC-3′) were designed according to partial rDNA sequence comprising 25S and 18S rRNA genes separated with nontranscribed spacer (GenBank accession No. X52575). They were used in all possible combinations with either of the two PisTR-A primers (F5/1 and F5/2). Reaction conditions included initial denaturation at 94°C for 3 min; 30 cycles of amplification of 30 s at 94°C, 50 s at 55°C, and 3 min at 72°C; followed by final extension for 10 min at 72°C. The reaction mix consisted of 1× PCR buffer, 0.2 mM dNTPs, 0.2 μM primers, 1.5 mM $MgCl_2$, 1.4 ng/μL genomic DNA, and 4 U/100 μL Taq polymerase (Promega). PCR was carried out in a PTC-200 Peltier Thermal Cycler (MJ Research, Waltham, Mass.), and the reaction products were analyzed on 1% agarose gels.

Results

Cloning and characterization of dispersed repetitive sequences

The genomic DNA library was constructed by random cloning of *Taq*I-digested fragments and probed with labeled pea DNA, to identify clones containing repetitive genomic sequences. The size of the library (6400 clones with an average insert size of 700 bp) ensured that it contained all repeats comprising at least 5×10^3 copies/1C with 99% probability (Ausubel et al. 1991). Colonies showing strong hybridization signals were selected for evaluation of insert sizes and for more precise estimation of their genomic abundance. From these, 32 clones with the strongest hybridization signals (relative to their lengths) were selected for sequencing and detailed analysis (Table 1). The clones ranged from 149 to 2961 bp in length and their copy numbers ranged from 10^3 to 4×10^4/1C. There were no completely identical sequences found; however, several partially homologous or overlapping clones were identified. Five groups of overlapping clones were assembled, having from two to five members (Table 1). Hybridization patterns obtained in Southern blot experiments revealed that all sequences had dispersed genomic organization (Figs. 1A and 1B). The sequence similarities of clones assigned to the same groups were also reflected by similar or identical Southern hybridization patterns, whereas the rest

© 2001 NRC Canada

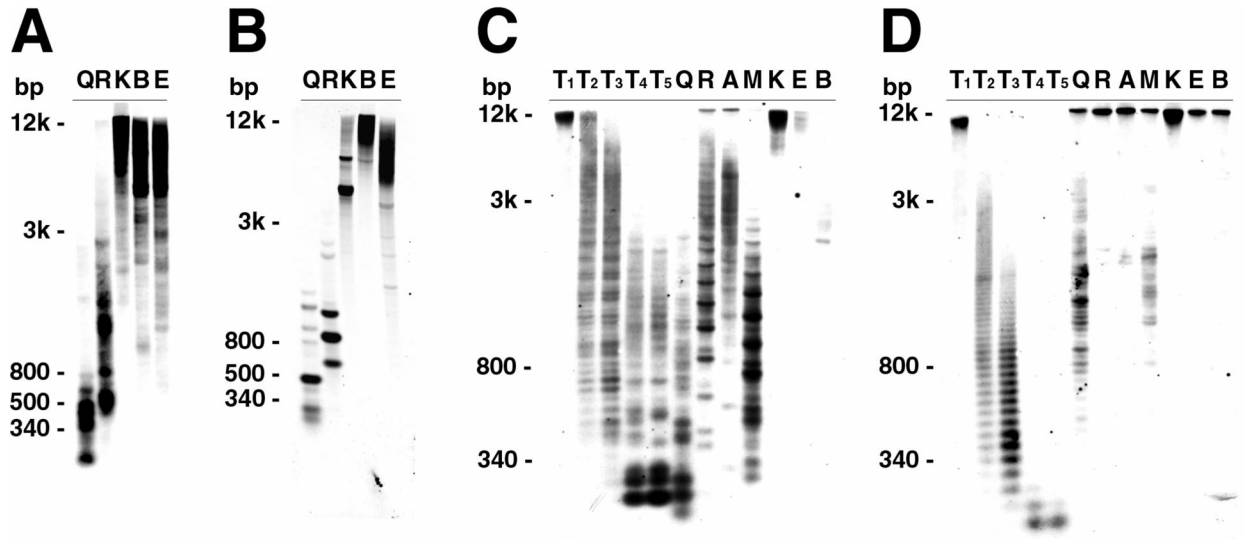
Table 1. Characterization of the clones isolated.

Clone	Length (bp)	AT content (%)	Copy no. (per 1C)	Proportion of the genome (%)	Homology to other clones (position/%)	GenBank accession No.
Psat1-1	468	56	33000	0.37	Psat1-5: 90–3/93	AF155741
Psat1-5	543	57	22000	0.27	Psat1-1: 4–89/93 Psat1-9: 90–542/93	AF155745
Psat1-9	659	57	15000	0.23	Psat1-5: 658–206/93	AF155749
Psat2-2	495	55	26000	0.30	Psat2-13: 267–468/50 Psat2-14: 6–316/60 Psat2-33: 1–495/91	AF155742
Psat2-13	502	62	8000	0.10	Psat2-2: 502–300/50 Psat2-14: 501–411/67 Psat2-33: 502–416/74	AF155752
Psat2-14	585	53	17000	0.24	Psat2-2: 239–581/60 Psat2-13: 472–559/67 Psat19: 1–23/78 Psat2-33: 227–578/57	AF155753
Psat2-33	512	56	29000	0.35	Psat2-2: 1–512/91 Psat2-13: 282–364/74 Psat2-14: 21–318/57	AF155770
Psat3-3	351	50	26000	0.21	Psat3-10: 1–351/64 Psat3-16: 4–149/64 Psat3-18: 4–149/84 Psat3-26: 1–219/53	AF155743
Psat3-10	469	49	29000	0.31	Psat3-3: 100–469/64 Psat3-16: 120–247/54 Psat3-18: 120–247/64 Psat3-26: 1–323/64	AF318289
Psat3-16	359	52	13000	0.11	Psat3-3: 186–332/ 64 Psat3-10: 186–332/54 Psat3-18: 1–359/80 Psat3-26: 1–174/55, 175–332/59	AF155754
Psat3-18	375	51	30000	0.27	Psat3-3: 199–341/84 Psat3-10: 199–341/64 Psat3-16: 1–375/80 Psat3-26: 1–192/60, 193–341/59	AF155756
Psat3-26	517	50	18000	0.22	Psat3-3: 242–1/53 Psat3-10: 359–1/64 Psat3-16: 517–340/55, 245–78/59 Psat3-18: 517–340/60, 245–78/59	AF155763
Psat4	957	60	16000	0.35		AF155744
Psat6	607	54	19000	0.27		AF155746
Psat7	469	55	39000	0.43		AF155747
Psat8	663	54	20000	0.31		AF155748
Psat11	899	57	8000	0.16		AF155750
Psat12	405	55	7000	0.06		AF155751
Psat17	542	58	5000	0.06		AF155755
Psat19	168	55	15000	0.06	Psat14: 148–168/78	AF155757
Psat20	325	55	6000	0.05		AF155758
Psat22	937	65	2000	0.04		AF155759
Psat23	212	57	16000	0.08		AF155760
Psat24	1134	60	13000	0.36		AF155761
Psat25	1023	58	5000	0.13		AF155762
Psat27-27	1168	60	8000	0.21	Psat27-29: 1168–2/88	AF155764
Psat27-29	2089	63	10000	0.5	Psat27-27: 1–1165/88	AF155766
Psat28	195	57	18000	0.08		AF155765
Psat30-30	2961	67	9000	0.65	Psat30-31: 1–2798/60	AF155767
Psat30-31	2591	66	10000	0.59	Psat30-30: 1–2504/60	AF155768
Psat32	490	59	10000	0.11		AF155769
Psat34	149	54	1000	0.002		AF155771
PisTR-A	211	61	20000	0.1		AF300839–AF300842
PisTR-B	50	72	10000	0.01		AF300830–AF300838

Note: Homology is given as nucleotide position of overlapping region / % of identity.

© 2001 NRC Canada

Fig. 1. Examples of Southern blots of clones showing dispersed or tandem genomic organization. *Pisum sativum* genomic DNA was digested with *Taq*I (Q), *Rsa*I (R), *Kpn*I (K), *Bam*HI (B), *Eco*RV (E), *Tru*9I (T1–T4 are partial digests and T5 is a complete digest), *Alu*I (A), and *Mbo*I (M) and probed with Psat3-3 (panel A), Psat32 (panel B), PisTR-A (panel C), or PisTR-B (panel D).



of the clones (16) that remained unassigned showed unique hybridization patterns (data not shown).

Sequence similarity searches of DNA and protein sequence databases revealed that 14 clones represented parts of retroelements or retroelement-like sequences, as judged from their homologies to reverse transcriptase – ribonuclease H (RT–RH), protease (PR), integrase (INT), gag (ORF (open reading frame) encoding the coat protein), or long terminal repeat (LTR) regions (Table 2). Five clones displayed homologies to unassigned genomic sequences from *Vicia faba*, *Medicago sativa*, and *P. sativum*, and the rest (13 clones) represented repetitive elements of unknown origin.

Clones with homology to retroelements

Five clones were identified as homologous to RT–RH domains at both DNA and protein levels (Table 2). The best homology score among all our clones (94.3% homology in 490 bp) was produced by the clone Psat32 with the previously sequenced retrotransposon *peabody* (GenBank accession No. AF083074; G.P. Volman, J. Shilo, H. Weinhouse, P. Ohana, and N. Benziman, unpublished data). The Psat32 sequence does not contain any stop codon and could be translated to protein in one reading frame. Southern blot hybridization to digested pea DNA produced several strong bands with little smear (Fig. 1B), which, together with well-conserved nucleotide sequences, suggests that this retrotransposon could have undergone recent amplification in the pea genome. In situ hybridization on pea mitotic chromosomes showed that Psat32 sequences were dispersed over all chromosomes, except for weaker or missing signals in secondary constrictions (Fig. 5D).

Three clones belonging to the Psat1 group matched corresponding RT–RH sequences from many organisms (Tables 1 and 2). Two additional clones (Psat12 and Psat28) could also be putatively assigned to the Psat1 group, based on Southern blot patterns similar to those of the Psat1 sequences and partial homology to the RT–RH and protease domains (Table 2).

Psat19 and clones of the Psat2 group, with the exclusion of Psat2-13, were homologous to the integrase domain, showing the best protein homology to the *Nicotiana tabacum* L. translated ORF lying upstream of the str 246N pseudogene (Froissard et al. 1994). They also exhibited significant protein homology with integrase-coding sequences of several retroelements (Table 2). Similar to the FISH experiments with the clone Psat32, the integrase sequences represented by the clone Psat19 were found to be uniformly distributed on all pea chromosomes (data not shown).

There were several clones homologous to less-conserved retroelement regions (gag, LTR); these included clone Psat6 (homology to the gag sequences of several plant retroelements) and Psat22 and Psat24 (LTR-like sequences; Table 2). We detected a potential primer binding site, consisting of a 14-bp region with 100% homology to the 3′ end of $tRNA_{ini}$, in Psat24 that supports the assignment of this clone to LTR sequences (Kumar and Bennetzen 1999). The clone Psat7 shared a short DNA homology (71% of 112 bp) with the pol region (ORF encoding PR, RT–RH, and INT) of the retrovirus-like element *Calypso* from *Glycine max* (GenBank accession No. AF186186; D.A. Wright and D.F. Voytas, unpublished data). However, since this sequence is not present in the pol regions of other retroelements, it is not clear whether it represents an integral part of or only an insertion of unrelated sequence into the *Calypso* pol region.

Dispersed repeats with no detectable homology to retroelements

There were several clones homologous to dispersed repeats from other legumes (Psat8) or to uncharacterized sequences surrounding several genes in *P. sativum* (Psat20, Psat25, Psat30-30, and Psat30-31). The rest of the clones (13) represented novel repeats with no detectable homology to known sequences. However, as revealed by sequence analysis, some of these clones contain interesting features. The Psat17 sequence includes a region of 54 bp made up mostly of adenines (80%) and resembling the poly(A) regions of

© 2001 NRC Canada

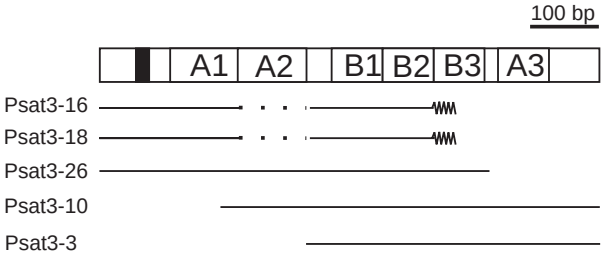
Table 2. Clones with homology to known sequences.

Clone	Description	DNA homology	Protein homology
Psat1-1, -5, -9	RT–RH	*Pisum sativum* (AJ002213, U25035), *Vicia narbonensis* (AJ391757, AJ391756), *Vicia melanops* (AJ391755, AJ391759), *Vicia faba* (M16855, M16858), *Zea diploperennis* (**X97604**), *Sorghum bicolor* (**AF098806**)	*Arabidopsis thaliana* (AAF79618), *Oryza sativa* (BAA92417), *Zea mays* (**T14595**), *Saccharomyces cerevisiae* (**NP_011624**), *Vicia faba* (**T12085**), *Ananas comosus* (**T07863**), *Sorghum bicolor* (**AAD22153**)
Psat2-2, -14, -33, Psat19	Integrase	*Vicia melanops* (AJ391774, AJ391775), *Gossypium barbadense* (AF060652), *Lens culinaris* (AJ401231), *Vicia faba* (M16857), *Nicotiana tabacum* (**X80830**), *Arabidopsis thaliana* (AC027665), *Vicia sativa* (AJ391773)	*Nicotiana tabacum* (**S47444**), *Arabidopsis thaliana* (AAD22283), *Beta vulgaris* (BAA99310.1), human immunodeficiency virus type 1 (**AAD44374**)
Psat6	Gag	*Medicago truncatula* (AL373727, AL377321), *Pisum sativum* (**AF083074**), *Zea diploperennis* (X87229), *Lilium henryi* (**X13886**), *Solanum tuberosum* (AF265664), *Hordeum vulgare* (Y14573), *Oryza sativa* (AB014741)	*Oryza sativa* (AAD27547.1), *Lilium henryi* (**1510387A**), *Sorghum bicolor* (AAD22157, AAD22153), *Oryza sativa* (BAA90795), *Arabidopsis thaliana* (T01862)
Psat7	Pol region	*Vicia narbonensis* (AJ391780), *Glycine max* (BE058976, **AF186186**)	None
Psat8	Repetitive element	*Glycine max* (AZ301674, AZ254370, AZ254366, AQ989396), *Vicia faba* (**M16856**), *Medicago sativa* (**L39961**), *Vicia narbonensis* (**AJ391771**), *Medicago truncatula* (AL366606)	None
Psat12	Protease	*Glycine max* (AI856184, AZ482492, AI966155, AI855678, AZ301954, AZ302022, AZ482490, AZ301357, AZ044859), *Medicago truncatula* (AW684517), *Homo sapiens* (AQ246751, AQ091390)	*Oryza sativa* (BAA96774, **BAA84457**), *Arabidopsis thaliana* (AAF79618)
Psat20, Psat25	Unknown	*Pisum sativum* (AF081457)	None
Psat22	LTR	*Glycine max* (AZ081679, **AF053008**, **U96295**), *Vicia faba* (AJ237842)	None
Psat24	LTR	*Oryza australiensis* (**D85597**, **D85598**), *Glycine max* (AZ044872)	None
Psat28	RT–RH	*Vicia faba* (M16858, M16851, M16848, M16854, M16847, M16849, M16852, M16850), *Lycopersicon esculentum* (AQ367541), *Oryza sativa* (AC022352)	*Oryza sativa* (BAA96774), *Arabidopsis thaliana* (AAD30632), *Drosophila buzzatii* (**CAB39733**)
Psat30-30, -31	Unknown	*Pisum sativum* (AF138704, X80007), *Vicia faba* (AJ243461)	None
Psat32	RT–RH	*Pisum sativum* (**AF083074**, AJ276900), *Sorghum bicolor* (**AF061282**), *Arabidopsis thaliana* (AL138640), *Oryza sativa* (AC078944), *Lilium henryi* (**X13886**), *Mus musculus* (AC080168, AC051621), *Hordeum vulgare* (**AJ002619**, **AJ002618**), *Ananas comosus* (**Y12432**)	*Sorghum bicolor* (**AF061282**), *Oryza sativa* (**AAD38284.1**), *Ananas comosus* (**T07863**), *Nicotiana tabacum* (**BAA75236.1**), *Hordeum vulgare* (**AB014746**)

Note: Only the sequences giving the best scores are listed (in descending order of their homology scores). The GenBank accession numbers of the entries that allowed the assignment of clones to individual retroelement domains are indicated in bold.

© 2001 NRC Canada

Fig. 2. Schematic representation of the alignment of Psat3 clones. Boxes with letters indicate the positions of A and B subrepeats; the microsatellite motif $(CA)_n$ is indicated by a black box. Missing regions are represented by dotted lines and nonhomologous regions by zig-zag lines.



non-LTR retroelements (LINEs (long interspersed elements) or SINEs (short interspersed elements)) (Kumar and Bennetzen 1999). The Psat3 group comprises five clones that can be ordered into a contig (Fig. 2). The region represented by the Psat3 clones contains two types of tandemly organized subrepeats (A and B) and a microsatellite ($(CA)_n$). Both subrepeats, as well as the microsatellite, differ among individual clones in the number of the monomer repetition, as depicted in Fig. 2. Subrepeat A is 90–100 bp long and is present as two full-length copies in Psat3-26 and only once in Psat3-10, Psat3-16, and Psat3-18. Moreover, a shorter region (65 bp) homologous to this subrepeat is located downstream from the B subrepeat in Psat3-3 and Psat3-10. Homology between the two A subrepeats in Psat3-26 and Psat3-10, and among these clones, ranges between 41–49 and 45–97%, respectively. Subrepeat B is 69 bp in length and is best conserved in Psat3-26, which contains three copies with a high mutual homology (86–94%). Three copies are also present in clones Psat3-3 and Psat3-10, but they are less conserved (51–66 and 71–73%, respectively). This is also the case for Psat3-16 and Psat3-18, which have only two copies, with homologies of 58 and 67%, respectively. As the B subrepeats were at least duplicated in all Psat3 clones, we were interested in whether they were amplified to longer tandem arrays in other copies of this sequence present in the pea genome. Therefore, a Southern blot of genomic DNA partially or completely digested with *Fok*I (its recognition site is present once in each B monomer in Psat3-26) was hybridized with labeled B subrepeats of Psat3-26. This resulted in a smear with several prominent bands but lacked the ladder pattern typical for long arrays of tandemly organized repeats (data not shown). Moreover, FISH experiments using this probe produced a dispersed labeling pattern similar to that of the retroelement sequences (Fig. 5E), thus confirming that the B subrepeats do not form large clusters in the pea genome.

Isolation and characterization of tandem repeats

Hybridization with Cot-1 DNA

Seventy clones from the *Taq*I genomic library known to contain at least moderately repeated sequences (based on their hybridization to total DNA) were reprobed with labeled pea Cot-1 DNA. The hybridization intensities of individual clones were compared with those obtained after hybridization with total genomic DNA, and the five clones showing the highest Cot-1 to genomic DNA signal ratio were selected for sequencing. Three of these clones—Psat35 (GenBank accession No. AF300844), Psat36 (AF300845), and Psat37 (AF300843)—were identified as homologs of clone Psat28, tandem subrepeats of pea nontranscribed spacer of ribosomal DNA, and chloroplast DNA sequences, respectively. One of the remaining two clones produced a ladder pattern on a Southern blot of pea DNA digested with various restrictases, suggesting a tandem genomic organization (Fig. 1C). Since this clone was only 49 bp in length, whereas the main ladder spacing observed on the blot was about 200 bp, it was likely that it represented only a part of the repeat monomer. Therefore, we attempted to isolate full-length clones using PCR with primers directed outward from the known sequence. The reaction resulted in several discrete bands with the expected (~200 bp) spacing (Fig. 4B, lane 1), and fragments corresponding to the repeat monomer were cloned and sequenced. The consensus sequence of this tandem repeat (named PisTR-A) was reconstructed from four sequenced clones (Fig. 3A). The sequence is 212 bp long and is present in 2×10^4 copies per haploid pea genome. Interestingly, FISH experiments using the PisTR-A probe resulted in partially dispersed signal with prominent labeling of NORs (nucleolar organizing regions). Although both NORs diplayed an accumulation of PisTR-A sequences, the signals on chromosome 7 were clearly stronger (Fig. 5C). To investigate its organization with respect to rRNA genes, six PCR primers directed towards rRNA genes or nontranscribed intergenic spacer (IGS) were designed (Fig. 4A) and used in all possible combinations with PisTR-A primers. Only a combination of the rDNA3 and F5/1 primers resulted in a product (~800 bp), suggesting the presence of PisTR-A in the 3′ direction from the 25S rRNA gene (Fig. 4B).

GSP-PCR

GSP-PCR was performed with pea genomic DNA digested with one of the 13 restriction endonucleases (either a 4- or a 6-bp cutter) listed in Materials and methods. Although all reactions produced a smear visible after agarose gel electrophoresis, only *Tru*9I produced a ladder of bands after redigestion of the PCR products with the same enzyme (data not shown). Following cloning into a plasmid vector, nine fragments corresponding to the repeat monomer, dimer, or tetramer were sequenced and assigned to a novel PisTR-B family. A consensus sequence derived from these clones was 50 bp long and contained a high proportion of A/T (72%) (Fig. 3B). Mutual homologies between 15 sequenced monomers were 50–90% (average 77%), and homologies to the consensus were 54–94%. Whereas only monomer and dimer fragments were detectable on Southern blots with genomic DNA completely digested with *Tru*9I, partially digested samples confirmed a regular ladder pattern with 50-bp spacing (Fig. 1D). We found sequence homology between PisTR-B and a family of *Vicia* satellite DNA, VicTR-B (Fig. 3C). This homology involves 40 bp of the PisTR-B sequence, including a AAATTTG motif that is also conserved in another *Vicia* tandem repeat family, VicTR-A (Macas et al. 2000).

The abundance of PisTR-B repeats was estimated to be 10^4/1C and, in contrast to PisTR-A sequences, they were localized as discrete bands on pea mitotic chromosomes using

© 2001 NRC Canada

Fig. 3. Consensus sequences of PisTR-A (A) and PisTR-B (B) repeats displayed as sequence logos. The logos were created using the method of Schneider and Stephens (1990), as implemented at http://www.bio.cam.ac.uk/cgi-bin/seqlogo/logo.cgi. The positions of primers F5/1 and F5/2 are indicated above the PisTR-A sequence. (C) Alignment of the PisTR-B and VicTR-B consensus sequences.



FISH. The PisTR-B probe produced mostly (sub)telomeric and pericentromeric signals whose patterns, together with chromosome morphology, allowed discrimination of all chromosome types within the pea karyotype (Fig. 5).

Distribution of pea repetitive sequences in other legume species

The distribution of individual sequences in other plant genomes was tested in 19 legume species and *A. thaliana* by quantitative dot blot hybridization. The results are summarized in Table 3. As expected, most of the pea dispersed repeats (especially those with homology to conserved retroelement sequences) were detected in other species, especially in *Vicia* spp., where in some cases, they exceeded 1000 copies/1C. On the other hand, no more than 1000 were detected, and only very rarely were 100–1000 copies of any pea sequence detected, in species other than *Vicia* spp.

In contrast to dispersed repeats, the tandemly organized sequences PisTR-A and -B were mostly limited to species of *Pisum*. PisTR-A repeats did not produce any signal when hybridized to other legumes or *A. thaliana*, whereas PisTR-B was detected in hundreds of copies in several species of *Vicia*. (Table 3). However, whether this signal reflected cross-hybridization of PisTR-B to highly abundant and partially homologous *Vicia* satellite sequences or was produced by low numbers of PisTR-B repeats remains to be tested.

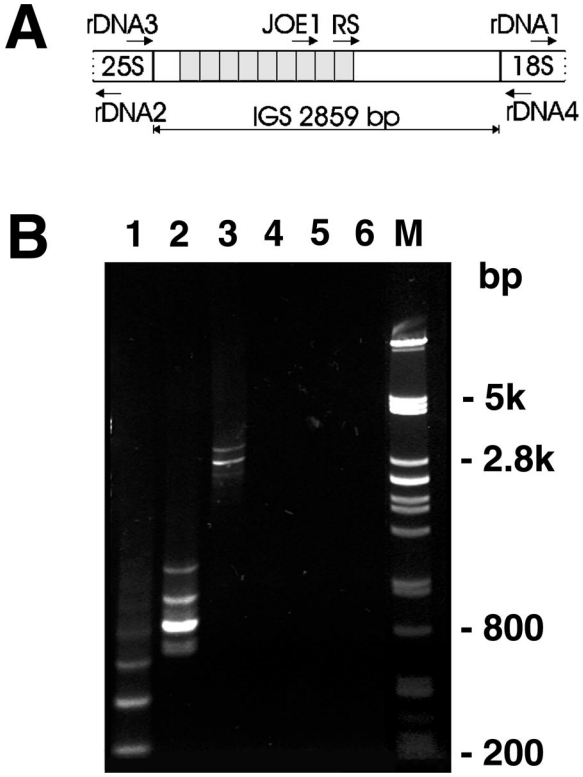
Discussion

The combination of techniques used in this work proved to be efficient for isolating repetitive sequences that differ in their genomic organization. Although, owing to the tremendous complexity of higher plant genomes, the set of obtained clones cannot be considered complete even for abundant repeats, it can be regarded as a representative sample of the composition of pea repetitive DNA. The isolated clones range in size from 10^3 to 4×10^4 copies/1C and comprise 4.5% of the pea genome. However, it should be noted that many of these clones are parts of various retrotransposons and that, therefore, the corresponding full-length elements probably span a much larger portion of the genome (2–10% in total, considering they are 3.2–13.5 bp long (Kumar and Bennetzen 1999)).

Copy numbers of newly isolated repeats were in the range predicted from reassociation studies of pea genomic DNA (Murray et al. 1978). Most of the clones represented dispersed repeats that are ubiquitous in higher plants and often make up the majority of their genomes (Smyth 1991;

© 2001 NRC Canada

Fig. 4. (A) A scheme of a *Pisum sativum* rDNA intergenic spacer, with the positions of the primers used for PCR in combination with PisTR-A primers. The 3′ end of the 25S rRNA gene and the 5′ end of the 18S rRNA gene are marked 25S and 18S, respectively; internal subrepeats present in the spacer are shaded (the drawing is based on GenBank sequence X52575). (B) An electrophoresed agarose gel (1%) of PCR products obtained using the following primer combinations: lane 1, F5/1 + F5/2 (PisTR-A specific primers); lane 2, F5/1 + rDNA3; lane 3, rDNA3 + rDNA4; lane 4, rDNA4 + F5/2; lane 5, F5/1 + JOE1; lane 6, F5/1 + RS; lane M, λ DNA digested with *Pst*I.



Bennetzen and Kellogg 1997; Bennetzen 1998; Kumar and Bennetzen 1999). Forty-four percent (14 of 32) of these clones were directly identified as retrotransposon sequences, whereas the rest showed very little or no homology to known elements. This can be explained by their origin from less-conserved parts of LTR-retrotransposon sequences, like LTRs (SanMiguel et al. 1998), or regions outside gag–pol coding sequences (Martinez-Izquierdo et al. 1997; Suoniemi et al. 1997; Chavanne et al. 1998; Marillonnet and Wessler 1998; Ohtsubo et al. 1999). Alternatively, they may represent fragments of non-LTR retrotransposons, like LINEs or SINEs (Okada et al. 1997; Noutoshi et al. 1998; Terai et al. 1998; Noma et al. 1999), or unknown or poorly defined elements like MITEs (miniature inverted-repeat transposable elements; Wessler 1998). This could be the case for the Psat17 clone, which contains poly(A) tracts typical of SINEs and LINEs (Kumar and Bennetzen 1999).

The majority of sequences identified as parts of retroelements showed homology to Ty3-*gypsy* retrotransposons (Table 2). The prevalence of Ty3-*gypsy*- over Ty1-*copia*-like sequences was also observed in several species of *Vicia*. (Nouzová et al. 2001) and in *Cicer* (Staginnus et al. 1999), so it may be a more general phenomenon. The uniform distribution of retrotransposon sequences on pea chromosomes resembles the dispersed patterns frequently observed in FISH experiments with other plant species (Sentry and Smyth 1985; Hagemann et al. 1993; Galasso et al. 1997; Heslop-Harrison et al. 1997; Schmidt and Heslop-Harrison 1998). The only regions on the pea chromosomes displaying no or very low signals were at secondary constrictions that contain rDNA and PisTR-A sequences (Fig. 5D).

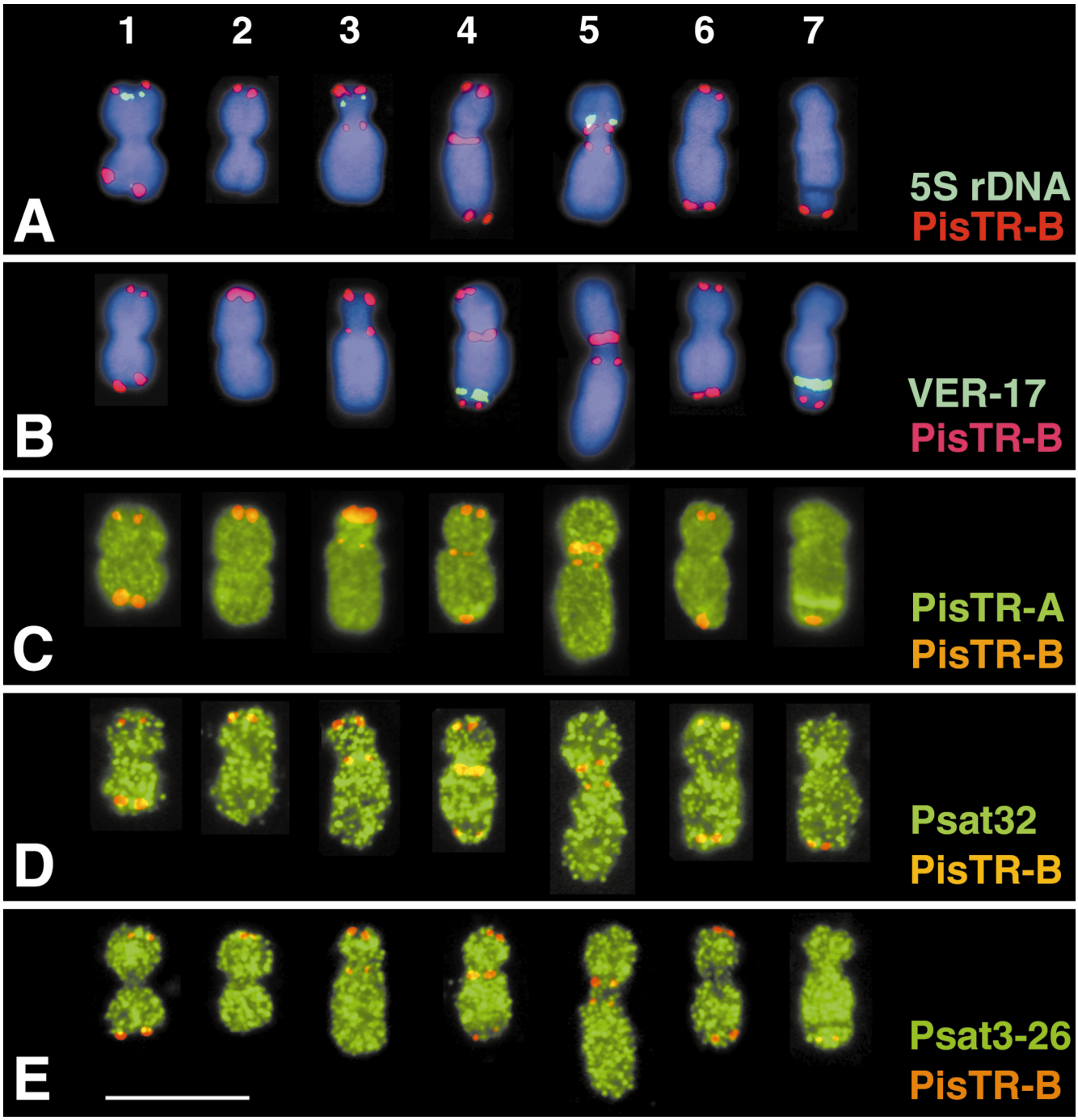
The two families of tandem repeats described in this study are not as abundant (copy numbers of 2×10^4 and 10^4/1C) as highly amplified satellites from related legume species (copy numbers of 10^6–10^7/1C) (Maggini et al. 1995; Macas et al. 2000). Although the presence of such satellite(s) in pea cannot be ruled out, it is unlikely, owing to the relatively small heterochromatic bands detected on pea chromosomes (Lamm 1981) and the absence of bands lacking a signal after FISH with dispersed repeats (Figs. 5C–5E). Such "negative" bands were observed, for example, at regions containing *Fok*I repeats in field bean and AbS1 repeats in *Anemone blanda* (Hagemann et al. 1993; Fuchs et al. 1994; Pearce et al. 1996).

The partially dispersed signal produced on chromosomes by the PisTR-A probe is atypical for abundant tandem repeats; however, similar results were described for TIII15 sequences in *V. faba* (Nouzová et al. 1999). In both cases, the head-to-tail orientation of repeat monomers was proved on a small scale (several kilobases) by Southern blot hybridizations (Fig. 1C; Nouzová et al. 1999); therefore, FISH patterns can suggest the presence of many relatively small clusters of these repeats dispersed in the genome. It has been shown that certain mobile elements contain arrays of tandemly repeated sequences of up to nine monomers (Monfort et al. 1995; Martinez-Izquierdo et al. 1997; Noma and Ohtsubo 2000). The existence of such elements containing corresponding sequences or their homologs could explain the dispersed patterns of PisTR-A and TIII15 sequences; however, among our clones, we did not find any that could support this hypothesis. An additional interesting finding was the concentration of PisTR-A sequences in the vicinity of rRNA genes (Fig. 5C). The results of PCR reactions (Fig. 4B, lane 2) suggest that PisTR-A and rDNA repeats are not extensively interspersed but that, at least in some cases, the PisTR-A sequences directly follow the 3′ end of the 25S rDNA. However, owing to the limitations of PCR with respect to amplification of longer fragments, some other technique (fiber-FISH or cloning) will have to be used to address this problem. A co-localization of a tandem repeat (CaSat1) and rRNA genes was recently reported in *Cicer* using FISH; however, the mutual arrangement of these sequences is not known (Staginnus et al. 1999).

PisTR-B sequences differ from PisTR-A sequences in many aspects, including their monomer size, A/T content, and chromosomal localization, so they can be considered to be unrelated repeat families. However, PisTR-B sequences share partial sequence homology and similar monomer length (50 and 38 bp, respectively) with *Vicia* VicTR-B repeats (Macas et al. 2000). A generally close relationship of *Vicia* and *Pisum* repeats was also emphasized by the se-

© 2001 NRC Canada

Fig. 5. FISH on isolated pea chromosomes. One chromosome of each type (n = 7) is shown and the probes used for double hybridizations are indicated with colors corresponding to the signals on the chromosomes. The PisTR-B probe was labeled directly with Alexa Fluor 568 (red); all other probes were biotin-labeled and visualized using avidin-fluorescein (green). Chromosomes in A and B were counterstained with DAPI (blue). The assignment of chromosome numbers to individual chromosome types was based on the distribution of 5S rDNA, 45S rDNA (probe Ver17), and chromosome morphology (Ellis et al. 1988; Simpson et al. 1990; Fuchs et al. 1998). Scale bar = 10 μm.



quence homologies of the dispersed repeats that were not assigned to any functional retroelement domain (hence not as conserved) (Table 2). These results are in agreement with interspecific hybridization experiments performed with *Vicia* repeats (Nouzová et al. 2001) and probably reflect a close evolutionary relationship between the two genera (Polhill and Raven 1981).

One of the welcome results of this work was finding the PisTR-B repeats, which are suitable for unambiguous discrimination of individual pea chromosomes using FISH.

© 2001 NRC Canada

Table 3. Distribution of pea repetitive sequences in other species.

	Pisum sativum	*Pisum elatius*	*Vicia faba faba*	*Vicia hybrida*	*Vicia lutea*	*Vicia melanops*	*Vicia pannonica*	*Vicia michauxii*	*Vicia peregrina*	*Vicia narbonensis*	*Vicia lathyroides*	*Vicia grandiflora*	*Vicia sativa*	*Vicia sepium*	*Vicia villosa*	*Vigna unquiculata*	*Cicer arietinum*	*Glycine max*	*Lupinus angustifolius*	*Phaseolus vulgaris*	*Arabidopsis thaliana*
Psat1-1*	++	++	+	+	+	++	+	++	+	++	+	+	+	+	–	–	–	–	–	–	–
Psat1-5*	++	++	+	++	+	++	+	+	+	+	+	+	–	+	–	–	–	–	+	–	–
Psat1-9*	++	++	+	+	–	+	+	+	+	++	–	+	–	+	–	–	–	–	–	–	–
Psat2-2*	++	++	++	++	++	++	++	++	++	++	++	++	++	++	++	–	–	–	–	–	–
Psat2-13	++	++	++	++	++	++	++	++	++	++	++	++	++	++	+	–	–	–	+	–	–
Psat2-14*	++	++	++	+	+	+	+	++	++	++	+	+	+	+	–	–	–	–	–	–	–
Psat2-33*	++	++	++	++	++	++	++	++	++	++	++	++	++	++	++	–	–	–	–	–	–
Psat3-3	++	++	+	+	+	++	++	+	+	+	+	+	+	+	+	–	–	–	–	+	–
Psat3-10	++	++	–	–	–	+	–	+	+	+	–	–	–	–	–	–	–	–	–	–	–
Psat3-16	++	++	+	–	–	+	–	+	+	+	–	+	–	–	–	–	–	–	–	–	–
Psat3-18	++	++	–	–	–	–	–	+	–	+	–	–	–	–	–	–	–	–	–	–	–
Psat3-26	++	++	–	–	–	+	–	+	–	+	–	–	–	–	–	–	–	–	–	–	–
Psat4	++	++	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Psat6*	++	++	+	+	+	+	+	++	++	+	–	+	+	+	–	–	–	–	–	–	–
Psat7*	++	++	+	+	+	+	+	++	++	++	+	++	+	+	–	–	–	–	–	–	–
Psat8	++	++	–	–	–	+	–	+	–	–	–	–	–	–	–	–	–	–	–	–	–
Psat11	++	++	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Psat12*	++	++	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Psat17	++	++	–	–	–	–	–	–	–	+	–	–	–	–	–	–	–	–	–	–	–
Psat19*	++	++	++	++	++	+	++	++	+	++	+	++	+	+	–	+	–	–	–	–	–
Psat20	++	++	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Psat22*	++	++	++	++	+	+	–	++	++	+	–	+	+	–	+	–	–	–	–	–	–
Psat23	++	++	+	+	–	+	+	–	–	–	–	–	–	–	–	–	–	+	–	–	–
Psat24*	++	++	++	++	++	++	+	++	++	++	+	++	+	++	–	–	–	–	–	–	–
Psat25	++	++	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Psat28*	++	++	++	++	+	++	++	++	++	++	+	++	+	+	++	–	–	+	–	–	–
Psat27-27	++	++	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Psat27-29	++	++	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Psat30-30	++	++	+	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Psat30-31	++	++	+	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Psat32*	++	++	+	+	+	++	++	++	++	++	+	++	++	+	+	–	–	–	–	–	–
Psat34	++	++	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
PisTR-A	++	++	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
PisTR-B	++	++	–	+	+	+	–	–	–	+	–	–	–	–	–	–	–	–	–	–	–

Note: Abundance is given in copy number per haploid genome: ++, $>10^3$; +, 10^2–10^3; –, $<10^2$. The clones with homology to retroelement sequences are indicated by asterisks.

Such probes are of great value for karyotype and chromosome-translocation studies (Fuchs et al. 1994; Brandes et al. 1995; Busch et al. 1996; Cuadrado and Jouve 1997; Kamstra et al. 1997) and for sorting similarly sized chromosomes using flow cytometry (Macas et al. 1995; Pich et al. 1995).

Acknowledgements

We thank Ms H. Štěpančíková and Ms O. Šonková for excellent technical assistance and Ms G. Lambert for assistance in preparing the manuscript. This work was supported by grants 521/00/0655 and 521/96/K117 from the Grant Agency of the Czech Republic.

References

Altschul, S.F., Madden, T.L., Schäffer, A.A., Zhang, J., Zhang, Z., Miller, W., and Lipman, D.J. 1997. Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. Nucleic Acids Res. **25**: 3389–3402.

Ausubel, F.M., Brent, R., Kingston, R.E., Moore, D.D., Seidman, J.G., Smith, J.A., and Struhl, K. 1991. Current protocols in molecular biology. John Wiley & Sons, New York.

Bennett, M.D., and Smith, J.B. 1976. Nuclear DNA amounts in angiosperms. Philos. Trans. R. Soc. Lond. B Biol. Sci. No. 274. pp. 227–274.

Bennett, M.D., and Leitch, I.J. 1997. Nuclear DNA amounts in angiosperms—583 new estimates. Ann. Bot. (London), **80**: 169–196.

Bennetzen, J.L. 1998. The structure and evolution of angiosperm nuclear genomes. Curr. Opin. Plant. Biol. **1**: 103–108.

Bennetzen, J.L., and Kellogg, E.A. 1997. Do plants have a one-way ticket to genomic obesity? Plant Cell, **9**: 1509–1514.

Brandes, A., Roder, M.S., and Ganal, M.W. 1995. Barley telomeres are associated with 2 different types of satellite DNA-sequences. Chromosome Res. **3**: 315–320.

Busch, W., Herrmann, R.G., and Hohmann, U. 1996. Repeated DNA sequences isolated by microdissection. II. Comparative analysis in *Hordeum vulgare* and *Triticum aestivum*. Theor. Appl. Genet. **93**: 164–171.

Chavanne, F., Zhang, D.X., Liaud, M.F., and Cerff, R. 1998. Structure and evolution of *Cyclops*: a novel giant retrotransposon of

© 2001 NRC Canada

the *Ty3/gypsy* family highly amplified in pea and other legume species. Plant Mol. Biol. **37**: 363–375.

Cuadrado, A., and Jouve, M. 1997. Distribution of highly repeated DNA sequences in species of the genus *Secale*. Genome, **40**: 309–317.

Dellaporta, S.L., Wood, J., and Hicks, J.B. 1983. A plant DNA minipreparation: version II. Plant Mol. Biol. Rep. **1**: 19–21.

Ellis, T.H.N., Lee, D., Thomas, C.M., Simpson, P.R., Cleary, W.G., Newman, M.A., and Burcham, K.W.G. 1988. 5S Ribosomal-RNA genes in *Pisum*—sequence, long-range and chromosomal organization. Mol. Gen. Genet. **214**: 333–342.

Ellis, T.H.N., Delseny, M., Lee, D., and Burcham, K.W.G. 1990. Methylated and undermethylated rDNA repeats are interspersed at random in 2 higher-plant species. Plant Mol. Biol. **14**: 73–80.

Flavell, A.J., Smith, D.B., and Kumar, A. 1992. Extreme heterogeneity of *Ty1-copia* group retrotransposons in plants. Mol. Gen. Genet. **231**: 233–242.

Flavell, A.J., Knox, M.R., Pearce, S.R., and Ellis, T.H.N. 1998. Retrotransposon-based insertion polymorphisms (RBIP) for high throughput marker analysis. Plant J. **16**: 643–650.

Froissard, D., Gough, C., Czernic, P., Schneider, M., Toppan, A., Roby, D., and Marco, Y. 1994. Structural organization of str 246C and str 246N, plant defense-related genes from *Nicotiana tabacum*. Plant Mol. Biol. **26**: 515–521.

Fuchs, J., Pich, U., Meister, A., and Schubert, I. 1994. Differentiation of field bean heterochromatin by *in situ* hybridization with a repeated *Fok*I sequence. Chromosome Res. **2**: 25–28.

Fuchs, J., Kuhne, M., and Schubert, I. 1998. Assignment of linkage groups to pea chromosomes after karyotyping and gene mapping by fluorescence *in situ* hybridization. Chromosoma (Berlin), **107**: 272–276.

Galasso, I., Harrison, G.E., Pignone, D., Brandes, A., and Heslop-Harrison, J.S. 1997. The distribution and organization of *Ty1-copia*-like retrotransposable elements in the genome of *Vigna unguiculata* (L.) Walp. (cowpea) and its relatives. Ann. Bot. **80**: 327–333.

Gualberti, G., Doležel, J., Macas, J., and Lucretti, S. 1996. Preparation of pea (*Pisum sativum* L.) chromosome and nucleus suspensions from single root tips. Theor. Appl. Genet. **92**: 744–751.

Hagemann, S., Scheer, B., and Schweizer, D. 1993. Repetitive sequences in the genome of *Anemone blanda*: identification of tandem arrays and of dispersed repeats. Chromosoma (Berlin), **102**: 312–324.

Heslop-Harrison, J.S., Brandes, A., Taketa, S., Schmidt, T., Vershinin, A.V., Alkhimova, E.G., Kamm, A., Doudrick, R.L., Schwarzacher, T., Katsiotis, A., Kubis, S., Kumar, A., Pearce, S.R., Flavell, A.J., and Harrison, G.E. 1997. The chromosomal distributions of *Ty1-copia* group retrotransposable elements in higher plants and their implications for genome evolution. Genetica (The Hague), **100**: 197–204.

Ingle, J., Timmis, J.N., and Sinclair, J. 1975. The relationship between satellite deoxyribonucleic acid, ribosomal ribonucleic acid gene redundancy, and genome size in plants. Plant Physiol. **55**: 496–501.

Kamstra, S.A., Kuipers, A.G.J., DeJeu, M.J., Ramanna, M.S., and Jacobsen, E. 1997. Physical localisation of repetitive DNA sequences in *Alstroemeria*: karyotyping of two species with species-specific and ribosomal DNA. Genome, **40**: 652–658.

Kato, H., Sriprasertsak, P., Seki, H., Ichinose, Y., Shiraishi, T., and Yamada, T. 1999. Functional analysis of retrotransposons in pea. Plant Cell Physiol. **40**: 933–941.

Kumar, A., and Bennetzen, J.L. 1999. Plant retrotransposons. Annu. Rev. Genet. **33**: 479–532.

Lamm, R. 1981. Giemsa C-banding and silver-staining for cytological studies in *Pisum*. Hereditas, **94**: 45–52.

Lee, D., Ellis, T.H.N., Turner, L., Hellens, R.P., and Cleary, W.G. 1990. A *copia*-like element in *Pisum* demonstrates the uses of dispersed repeated sequences in genetic analysis. Plant Mol. Biol. **15**: 707–722.

Leitch, A.R., Schwarzacher, T., Jackson, D., and Leitch, I.J. 1994. *In situ* hybridization. BIOS Scientific Publishers Ltd., Oxford.

Macas, J., Doležel, J., Gualberti, G., Pich, U., Schubert, I., and Lucretti, S. 1995. Primer-induced labeling of pea and field bean chromosomes *in situ* and in suspension. BioTechniques, **19**: 402–408.

Macas, J., Požárková, D., Navrátilová, A., Nouzová, M., and Neumann, P. 2000. Two new families of tandem repeats isolated from genus *Vicia* using genomic self-priming PCR. Mol. Gen. Genet. **263**: 741–751.

Maggini, F., Dovidio, R., Gelati, M.T., Frediani, M., Cremonini, R., Ceccarelli, M., Minelli, S., and Cionini, P.G. 1995. *Fok*I DNA repeats in the genome of *Vicia faba*: species specificity, structure, redundancy modulation, and nuclear organization. Genome, **38**: 1255–1261.

Marillonnet, S., and Wessler, S.R. 1998. Extreme structural heterogeneity among the members of a maize retrotransposon family. Genetics, **150**: 1245–1256.

Martinez-Izquierdo, J.A., Garcia-Martinez, J., and Vicient, C.M. 1997. What makes Grande1 retrotransposon different? Genetica (The Hague), **100**: 15–28.

Monfort, A., Vicient, C.M., Raz, R., Puigdomenech, P., and Martinez-Izquierdo, J.A. 1995. Molecular analysis of a putative transposable retroelement from the *Zea* genus with internal clusters of tandem repeats. DNA Res. **2**: 255–261.

Murray, M.G., Cuellar, R.E., and Thompson, W.F. 1978. DNA sequence organization in the pea genome. Biochemistry, **17**: 5781–5790.

Murray, M.G., Peters, D.L., and Thompson, W.F. 1981. Ancient repeated sequences in the pea and mung bean genomes and implications for genome evolution. J. Mol. Evol. **17**: 31–42.

Noma, K., and Ohtsubo, E. 2000. Tnat1 and Tnat2 from *Arabidopsis thaliana*: novel transposable elements with tandem repeat sequences. DNA Res. **7**: 1–7.

Noma, K., Ohtsubo, E., and Ohtsubo, H. 1999. Non-LTR retrotransposons (LINEs) as ubiquitous components of plant genomes. Mol. Gen. Genet. **261**: 71–79.

Noutoshi, Y., Arai, R., Fujie, M., and Yamada, T. 1998. Structure of the *Chlorella Zepp* retrotransposon: nested *Zepp* clusters in the genome. Mol. Gen. Genet. **259**: 256–263.

Nouzová, M., Kubaláková, M., Doleželová, M., Koblížková, A., Neumann, P., Doležel, J., and Macas, J. 1999. Cloning and characterization of new repetitive sequences in field bean (*Vicia faba* L.). Ann. Bot. (London), **83**: 535–541.

Nouzová, M., Neumann, P., Navrátilová, A., Galbraith, D.W., and Macas, J. 2001. Microarray-based survey of repetitive genomic sequences in *Vicia* spp. Plant Mol. Biol. **45**: 229–244.

Ohtsubo, H., Kumekawa, N., and Ohtsubo, E. 1999. *RIRE2*, a novel *gypsy*-type retrotransposon from rice. Genes Genet. Syst. **74**: 83–91.

Okada, N., Hamada, M., Ogiwara, I., and Ohshima, K. 1997. SINEs and LINEs share common 3′ sequences: a review. Gene (Amst.), **205**: 229–243.

Pearce, S.R., Harrison, G., Li, D.T., Heslop-Harrison, J.S., Kumar, A., and Flavell, A.J. 1996. The *Ty1-copia* group retrotransposons in *Vicia* species: copy number, sequence heterogeneity and chromosomal localization. Mol. Gen. Genet. **250**: 305–315.

Pearce, S.R., Knox, M., Ellis, T.H.N., Flavell, A.J., and Kumar, A. 2000. Pea *Ty1-copia* group retrotransposons: transpositional ac-

© 2001 NRC Canada

tivity and use as markers to study genetic diversity in *Pisum*. Mol. Gen. Genet. **263**: 898–907.

Pearson, W.R., and Lipman, D.J. 1988. Improved tools for biological sequence comparison. Proc. Natl. Acad. Sci. U.S.A. **85**: 2444–2448.

Pich, U., Meister, A., Macas, J., Doležel, J., Lucretti, S., and Schubert, I. 1995. Primed *in situ* labeling facilitates flow sorting of similar sized chromosomes. Plant J. **7**: 1039–1044.

Polhill, R.M., and Raven, P.H. 1981. Papilionoideae. *In* Advances in legume systematics. Part 1. *Edited by* R.M. Polhill and P.H. Raven. Royal Botanic Gardens, Kew. pp. 191–409.

Richards, E.J., and Ausubel, F.M. 1988. Isolation of a higher eukaryotic telomere from *Arabidopsis thaliana*. Cell, **53**: 127–136.

Sanger, F., Nicklen, D., and Coulson, A.R. 1977. DNA sequencing with chain terminating inhibitors. Proc. Natl. Acad. Sci. U.S.A. **74**: 5463–5467.

SanMiguel, P., Gaut, B.S., Tikhonov, A., Nakajima, Y., and Bennetzen, J.L. 1998. The paleontology of intergene retrotransposons of maize. Nat. Genet. **20**: 43–45.

Schmidt, T., and Heslop-Harrison, J.S. 1998. Genomes, genes and junk: the large-scale organization of plant chromosomes. Trends Plant Sci. **3**: 195–199.

Schneider, T.D., and Stephens, R.M. 1990. Sequence logos: a new way to display consensus sequences. Nucleic Acids Res. **18**: 6097–6100.

Sentry, J.W., and Smyth, D.R. 1985. A family of repeated sequences dispersed through the genome of *Lilium henryi*. Chromosoma (Berlin), **92**: 149–155.

Simpson, P.R., Newman, M.A., Davies, D.R., Ellis, T.H.N., Matthews, P.M., and Lee, D. 1990. Identification of translocations in pea by in situ hybridization with chromosome-specific DNA probes. Genome, **33**: 745–749.

Smyth, D.R. 1991. Dispersed repeats in plant genomes. Chromosoma (Berlin), **100**: 355–359.

Staginnus, C., Winter, P., Desel, C., Schmidt, T., and Kahl, G. 1999. Molecular structure and chromosomal localization of major repetitive DNA families in the chickpea (*Cicer arietinum* L.) genome. Plant Mol. Biol. **39**: 1037–1050.

Suoniemi, A., Schmidt, D., and Schulman, A.H. 1997. *BARE*-1 insertion site preferences and evolutionary conservation of RNA and cDNA processing sites. Genetica (The Hague), **100**: 219–230.

Suoniemi, A., Tanskanen, J., and Schulman, A.H. 1998. *Gypsy*-like retrotransposons are widespread in the plant kingdom. Plant J. **13**: 699–705.

Terai, Y., Takahashi, K., and Okada, N. 1998. SINE cousins: the 3′-end tails of the two oldest and distantly related families of SINEs are descended from the 3′ ends of LINEs with the same genealogical origin. Mol. Biol. Evol. **15**: 1460–1471.

Vershinin, A.V., and Ellis, T.H.N. 1999. Heterogeneity of the internal structure of *PDR1*, a family of *Ty1/copia*-like retrotransposons in pea. Mol. Gen. Genet. **262**: 703–713.

Wessler, S.R. 1998. Transposable elements associated with normal plant genes. Physiol. Plant. **103**: 581–586.

Yakura, K., and Tanifuji, S. 1983. Molecular cloning and restriction analysis of *Eco*RI fragments of *Vicia faba* rDNA. Plant Cell Physiol. **24**: 1327–1330.

© 2001 NRC Canada